

CSCI 264 — Assembly Language
Lab 3
Pointers and Dynamic Memory in C

Write a C program, `stats.c`, which reads a count `n`, then `n` integers, and prints the values, their minimum, maximum, and average. It then swaps the smallest and largest values and prints them again.

Store the values in an array allocated with `malloc`, since the size is not known until the program runs. Check the value `malloc` returns for `NULL`, and `free` the array before the program ends. You may assume `n` is at least 1.

Functions

You must write each of these functions:

```
double average(int *arr, int n);
void min_max(int *arr, int n, int *min, int *max);
void swap(int *a, int *b);
```

- `average` finds the average of the values in the array. Note the values should be integers, but the average should be a float or double.
- `min_max` stores the *position* of the smallest element in `*min` and the position of the largest in `*max`. It does not return anything; `main` gets both answers through the pointer parameters. `main` needs the positions, not just the values, so it can swap those two elements.
- `swap` exchanges the two values that `a` and `b` point to. Call it once from `main` to exchange the smallest and largest values of the array.

Memory Addresses

Print the following memory addresses inside the functions using `%p`:

`main()`

- `n`
- the array: both `&arr` and `arr`
- `min` and `max`

`min_max()`

- the `arr` parameter: both `&arr` and `arr`
- the `min` parameter: both `&min` and `min`

`average()`

- the `arr` parameter: both `&arr` and `arr`

`swap()`

- your local variable holding the saved value (`temp` in the output below), once it is assigned
- `a` and `b`

Label each line with the function name and what is printed, lined up like this:

```
printf("[main]    &n    = %p\n", &n);
printf("[min_max] arr    = %p\n", arr);
```

Output

A run of your program should look like this, with real addresses in place of `0x...`:

```
How many values? 5
[main]    &n    = 0x...
[main]    &arr   = 0x...
[main]    arr    = 0x...
[main]    &min   = 0x...
[main]    &max   = 0x...
Enter 5 values: 4 -2 9 0 7
Values:    4 -2 9 0 7
[min_max] &arr   = 0x...
[min_max] arr    = 0x...
[min_max] min     = 0x...
[min_max] &min    = 0x...
Min:       -2
Max:       9
[average] &arr   = 0x...
[average] arr    = 0x...
Average:   3.60
[swap]    &temp  = 0x...
[swap]    a      = 0x...
[swap]    b      = 0x...
Swapped:   4 9 -2 0 7
```

Test case:

n	Values	Min	Max	Average	Swapped
5	4 -2 9 0 7	-2	9	3.60	4 9 -2 0 7

Submission

Submit your source file on Canvas by Tuesday, September 22 and demo the program to Professor West by the end of the next lab session. Put your name in a comment at the top of the source file. If a classmate helped you put their name in the comment. Disclose any use of Generative AI in a comment at the top.